

abhishek.sen

- [About](#)
- [Experience](#)
- [Projects](#)
- [Skills](#)
- [Achievements](#)
- [Contact](#)

Open to opportunities

Abhishek Sen

Robotics & Electronics engineer from IIITDM Kancheepuram. I build autonomous systems — from GPS-less Mars drones to ROS2 navigation stacks and Edge AI pipelines on Jetson hardware.

[View My Work ↓](#) [Download Resume](#)

4+

Internships

4+

Drone Projects

8.45

CGPA



Robotics Engineer @ RobotSpace

Who I am

About Me



Abhishek Sen

EC22B1098 · IIITDM Kancheepuram

Chennai, India

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+91-8269856166

B.Tech Electronics & Comm.

I'm a Robotics Engineer specialising in autonomous systems, ROS2 navigation, embedded hardware, and Edge AI. Currently working at **RobotSpace, Hyderabad**, building autonomous mobile robot (AMR) platforms for real-world industrial deployment.

Previously a **Project Assistant at IIT Madras** (GNC Lab), where I implemented a proprietary path-planning system using the Generalized Shape Expansion (GSE) algorithm — proven to significantly outperform traditional RRT* in speed and accuracy — deployed on NVIDIA Jetson Orin Nano with Livox 4D LiDAR.

Former Team Captain of **Aerobotics Team Astra** — led the team to national competitions including ISRO Robotics Challenge, SAE Aerothon, and IIT Roorkee Cognizance. Appreciated at National Space Day, New Delhi, by ISRO Chairman Dr. S. Somnath for building a GPS-less autonomous Mars analogue drone.

Degree

B.Tech (ECE)

CGPA

8.45 / 10

Batch

2022 - 2026

Status

● Graduating 2026

Career

Experience

Robotics Engineer

Jun 2026 - Present

RobotSpace · Hyderabad

- Designing and developing robotic systems for real-world deployment scenarios, applying expertise in ROS2, navigation and embedded systems.
- Working on autonomous mobile robot (AMR) platforms and intelligent control systems for industrial applications.

- Integrating sensors, actuators, and AI inference pipelines onto robot platforms for robust autonomous operation.

ROS2 Autonomous Robots Embedded Systems Hyderabad

Project Assistant

Oct 2025 – May 2026

Guidance, Navigation & Control Lab · IIT Madras

- Implemented a proprietary path-planning system using the Generalized Shape Expansion (GSE) algorithm in Gazebo simulation and real-time hardware via ROS2 & MAVROS.
- Deployed the navigation stack on an NVIDIA Jetson Orin Nano with a Livox 4D LiDAR for high-fidelity real-time data collection.
- Demonstrated GSE significantly outperforms traditional RRT* in obstacle avoidance speed & accuracy.
- Implemented sensor fusion pipelines (Kalman & Particle Filters) integrating IMU and 4D LiDAR data for precise state estimation.

ROS2 Humble SLAM MAVROS Jetson Orin Nano Sensor Fusion Gazebo

Hardware R&D Intern

May – Sep 2025

32Mins Digital Consultancy Services · IIT Madras Research Park

- Worked on an AI-driven learning partner device to enhance student engagement during online classes.
- Developed dual-camera calibration on Texas Instruments TDA4VM board and wrote a custom camera driver deployed in Docker.
- Improved user interaction and feedback loop to personalise learning using AI models.

Embedded Systems Edge AI TDA4VM Docker Image Processing

Project Intern

May – Jul 2024

Autonomous Systems Laboratory

- Developed an autonomous robot capable of navigating complex environments using LiDAR, SLAM, and Computer Vision — both simulation and real-time.

- Equipped the robot with real-time obstacle detection and avoidance for safe and efficient navigation.
- Implemented the GMapping algorithm to generate precise maps of the environment.

Computer Vision ROS2 SLAM GMapping

What I've built

Personal Projects

Autonomous Docking & Visual Servoing Robot

ROS2 package for autonomous docking using HSV-based computer vision to detect and track targets in Gazebo. Fuses monocular camera (bearing) with LiDAR LaserScan (distance) for precise target localization with a proportional controller.

ROS2 OpenCV Sensor Fusion

[GitHub →](#)

Robust Navigation & Trajectory Control

Custom lightweight navigation stack for TurtleBot3. Uses B-Spline Trajectory Smoothing for C^2 continuous paths and a Pure Pursuit Controller with Sequential Index Search for 20Hz high-frequency control in unstructured environments.

ROS2 Control Theory Python

[GitHub →](#)

Autonomous Drone — Martian Terrain Analysis

GPS-less autonomous drone for the ISRO Robotics Challenge. Uses Intel RealSense Depth Camera SLAM for mapping, YOLOv8 for object/boundary detection, and ROS2/MAVROS to convert vision data into real-time local position estimates.

YOLOv8 MAVROS SLAM

[GitHub →](#)

Autonomous Surveillance & Object Tracking Drone

Fully custom-designed autonomous drone leveraging YOLOv8 for real-time object detection and tracking during autonomous flight missions. Complete hardware and

software stack designed from scratch with ROS2 integration.

YOLOv8 ROS2 ML

[GitHub →](#)

Tech Stack

Skills & Tools

Robotics & Navigation

ROS2 Humble MAVROS SLAM GMapping ArduPilot Gazebo RViz2 Path Planning

AI / ML / Vision

YOLOv8 PyTorch TensorFlow OpenCV ResNet-18 Keras Scikit-Learn Edge AI

⚙️ Hardware & Embedded

Jetson Orin Nano Raspberry Pi Arduino ESP32 TDA4VM PCB Design Altium EasyEDA

Languages & Tools

Python C / C++ JavaScript HTML + CSS Docker Git / GitHub NumPy SciPy

Recognition

Achievements & Events

✈️ SAE Aerothon ISRO Robotics IIT Roorkee

National Drone Design Competition

SAE Aerothon

Society of Automotive Engineers India

India's premier annual UAV competition. Teams design, build, and fly autonomous drones through multiple challenge rounds — from detailed design reports and technical presentations to actual flight missions with waypoint navigation, payload drop, and precision landing. Team Astra competed against 100+ teams nationally in 2024 and 2025.

100+

Teams Competing

2

Years Participated

AIR 8

Best Ranking

Key Achievements

AIR 8

All India Rank 8 — Detailed Design Report & Presentation Round, SAE Aerothon 2025

Winner

Challenger Round Winner — SAE Aerothon 2024

AIR 21

All India Rank 21 — Report Documentation Round, SAE Aerothon 2024

Design

Designed fully autonomous fixed-wing UAV with precision navigation and payload delivery systems

Event Photographs

← 1 / 1 →



Mars Analogue Mission · National Finalist

ISRO Robotics Challenge 2025

Indian Space Research Organisation (IRoC-U 2025)

ISRO's prestigious national competition challenging students to build autonomous drones for simulated Martian terrain — no GPS allowed. Team Astra built a fully autonomous drone using Intel RealSense SLAM, YOLOv8 vision, and ROS2/MAVROS stack. Selected as top finalist and personally appreciated by ISRO Chairman S. Somnath and Indian astronauts at National Space Day, New Delhi.

Top

Finalist

0 GPS

Autonomous Flight

ISRO

Appreciation

Key Achievements

Finalist

Top finalist in ISRO Robotics Challenge 2025 (IRoC-U) — competing against top national teams

Chairman

Personally appreciated by ISRO Chairman Dr. S. Somnath at National Space Day, Bharat Mandapam, New Delhi

Astronaut

Demonstrated the drone to Indian astronaut Angad Pratap and senior ISRO scientists

SLAM

Built GPS-less autonomous drone with RealSense SLAM, YOLOv8 object detection & ROS2 for Martian terrain

Event Photographs



Team Astra at ISRO Space Exhibition — Chandrayaan & ISRO rocket models, National Space Day

DIGITAL TARA



GALAXEYE



MISSION DRISHTI

World's First SyncFused with SAR



Smart

HS | HES | IoT | SAR
Sensor Convergence



ISRO Chairman Dr. S. Somnath visited the Team Astra stall — Bharat Mandapam, New Delhi



Demonstrating the autonomous drone to Indian astronaut Angad Pratap at National Space Day



Team with defence personnel at the National Space Day expo — Bharat Mandapam, New Delhi



Live drone presentation at Bharat Mandapam — discussing the GPS-less autonomous navigation system

IRoC-U 2025 Final Field Round

IIT Roorkee — IRoC-U 2025

ISRO Robotics Challenge Final Field Round · IIT Roorkee Campus

The IRoC-U 2025 Final Field Round was hosted at IIT Roorkee, requiring teams to fly their autonomous drone in a real outdoor Martian analogue environment. Teams had to complete both manual and fully autonomous flight missions. Only the technically strongest teams could complete both rounds — and Team Astra was one of them.

Top 4

Teams Completed

2

Mission Rounds

IIT

Roorkee

Key Achievements

Top 4

One of only four teams to complete both manual and autonomous drone missions at IRoC-U Final Field Round

Auto

Successfully completed fully autonomous waypoint navigation mission under real competition conditions

Custom UAV

Competed with 100% custom-built drone — all hardware and autonomous software built from scratch

IIT Roorkee

Represented IIITDM Kancheepuram at the prestigious IRoC-U 2025 Final Field Round at IIT Roorkee campus

Event Photographs



Leadership

Positions & Activities



Team Captain

Aerobotics Team Astra

Jul 2023 – Jan 2026

Led the team in national drone competitions. Developed skills in PCB design, team management, Arduino IDE, and wireless communication using MAVLink protocol.

Deep expertise in flight controllers and propulsion systems.

MAVLink PCB Design Team Lead

NCC Cadet

National Cadet Corps

2023 – 2025

Active NCC Cadet, developing discipline, leadership and teamwork through structured training programmes.

Leadership Discipline

Coordinator

Dramatics Cultural Club

2023 – 2025

Coordinated cultural events and dramatic performances, honing communication and event management skills alongside technical studies.

Communication Event Mgmt

Get in touch

Contact

Let's build something great together.

I'm actively looking for full-time roles and collaborations in Robotics, Autonomous Systems, and Edge AI. Whether it's a research project or industry role — I'd love to

hear from you.

[Email](#)

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[GitHub](#)

github.com/abhishek-sen

[LinkedIn](#)

[in/abhishek-sen-aab105259](https://in.linkedin.com/in/abhishek-sen-aab105259)

Education

Bachelor of Technology — Electronics & Communication

Indian Institute of Information Technology Design and Manufacturing,
Kancheepuram

CGPA: **8.45**

Batch: **2022 - 2026**

Roll: **EC22B1098**

Relevant Coursework: Data Structures & Algorithms · OOP · Computer Vision ·
Embedded Systems

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